SSC MTS 2024 question paper with answers

22 October 2024 · Shift 2

80 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 10 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 15 questions

Q.1

Rajeev saves 75% of his monthly income. If his monthly expenditure is ₹37,500, then his monthly saving (in ₹) is:

(A) 1,11,500
(B) 1,10,500
(C) 1,20,250
(D) 1,12,500

Q.2

Find the simple interest (in ₹) on ₹3500 as a sum borrowed at 3% per year rate of interest for 5 years.

(A) 525
(B) 465
(C) 485
(D) 565

Q.3

Water in a conical flask with radius 7 cm and height 48 cm is poured into a cylindrical flask with radius 14 cm. The water level in the cylindrical flask is:

(A) 3 cm
(B) 2 cm
(C) 5 cm
(D) 4 cm

Q.4

By selling an article for ₹19, a dealer makes a profit of 10%. By how much should he increase his selling price (in ₹, rounded off to two decimal place) so as to make a profit of 40%?

(A) 6.18
(B) 8.18
(C) 7.18
(D) 5.18

Q.5

Raj prepares a 60 gm mixture by combining two ingredients X and Y. The cost of ingredient X is ₹80 per 5 gm, and the cost of ingredient Y is ₹80 per 10 gm. Ingredients X and Y are mixed in a manner that the cost of the resulting mixture is ₹14 per gm. What is the quantity of ingredient X (in gm) in the mixture?

(A) 30
(B) 45
(C) 15
(D) 60

Q.6

In a scheme, a shopkeeper offers 4% discount if you buy three or more products. Ritu buys three bottles of guava juice for ₹330. If the shopkeeper still makes a profit of 10%, find the profit he would make if had sold it without discount (rounded off to 2 decimal places).

(A) 15%
(B) 15.74%
(C) 15.28%
(D) 14.58%

Q.9

A person has to cover a distance of 8 km in 1 hour. If he covers one-fourth of the distance in one-third of the total time, then what should his speed (in km/h) be to cover the remaining distance in the remaining time so that the person reaches the destination exactly on time?

(A) 9
(B) 8
(C) 7
(D) 6

Q.10

Which of the following options is equal to the LCM of 12, 36, 72, 144, 288 and 576?

(A) 72
(B) 288
(C) 144
(D) 576

Q.12

The average height of 50 students in a class was 105 cm. A group of five students with an average height of 95 cm were admitted to this class. What is the average height (in cm) of the class of 55 students? (correct to the nearest integer)

(A) 102
(B) 100
(C) 105
(D) 104

Q.13

Walking at a speed of 5.4 km/h, Rohit takes 40 minutes to reach his school from his home. What time will he take if he walks 8 km/h?

(A) 23 minutes
(B) 24 minutes
(C) 27 minutes
(D) 25 minutes

Q.14

A hemispherical bowl of internal radius 24 cm contains liquid. The liquid is to be filled into cylindrical container of diameter 8 cm and height 12 cm. What is the number of containers that are necessary to empty the bowl?

(A) 48
(B) 45
(C) 50
(D) 39

Q.15

Find the simple interest (in ₹) on ₹3500 as a sum borrowed at 5% per year rate of interest for 3 years.

(A) 565
(B) 485
(C) 465
(D) 525

Q.17

15 kg rice costing ₹60 per kg and 8 kg rice costing ₹55 per kg are mixed with 7 kg rice costing ₹40 per kg. Find the average per kg price of mixed rice.

(A) ₹48
(B) ₹54
(C) ₹56
(D) ₹45

Q.18

A tradesman marks his goods 25% above the cost price and allows his customers 12% reduction on their bills. What percentage profit does he make?

(A) 10%
(B) 11%
(C) 9%
(D) 12%

Q.19

A earns ₹88 per hour and works for 10 hours. B earns ₹95 per hour and works for 8 hours. The ratio of per day wages of A to B is:

(A) 19 : 22
(B) 21 : 19
(C) 22 : 19
(D) 19 : 21

Reasoning Ability and Problem Solving 16 questions

Q.1

In a certain code language, 'WANT' is coded as '8163' and 'THAN' is coded as '3468'. What is the code for 'H' in that language?

(A) 1
(B) 3
(C) 8
(D) 4

Q.2

In a certain code language, 'PAID' is coded as '3195' and 'GRAB' is coded as '2346'.What is the code for 'A' in the given code language?

(A) 1
(B) 3
(C) 2
(D) 4

Q.3

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some plants are spices. No spice is a lentil. Conclusions: (I) Some lentils are plants. (II) Some spices are plants.

(A) Only conclusion (II) follows
(B) Both conclusions (I) and (II) follow
(C) Only conclusion (I) follows
(D) Neither conclusion (I) nor (II) follows

Q.5

What should come in place of the question mark (?) in the given series based on the English alphabetical order? ADG, JMP, SVY, BEH, ?

(A) LNQ
(B) KMO
(C) LMO
(D) KNQ

Q.6

DZRG is related to HDOD in a certain way based on the English alphabetical order. In the same way, LHLA is related to PLIX. To which of the following is TPFU related, following the same logic?

(A) YUBQ
(B) WSCQ
(C) XTCR
(D) YTDS

Q.7

DBCA is related to HDFB in a certain way based on the English alphabetical order. In the same way, BACD is related to DBFH. To which of the following is FDAC related, following the same logic?

(A) LHBF
(B) LEFG
(C) LPTG
(D) LBGH

Q.8

21 is related to 189 following a certain logic. Following the same logic, 35 is related to 315. To which of the following is 43 related, following the same logic? (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding / subtracting / multiplying, etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 385
(B) 387
(C) 397
(D) 395

Q.9

Seven people R, S, T, D, E, F and G are sitting around a circular table, facing the centre (but not necessarily in the same order). R is sitting third to the right of D. Only one person is sitting between G and E when counted from the left of G. S is sitting to the immediate right of T. Only one person is sitting between G and T when counted from the right of G. Who is sitting third to the right of F?

(A) T
(B) S
(C) G
(D) D

Q.10

Six friends - A, B, C, D, E and F are sitting around a circular table facing away from the centre (but not necessarily in the same order). F is an immediate neighbour of both B and C. D is sitting second to the right of B. A is sitting to the immediate right of D. Who among the following is sitting to the second to the left of F?

(A) B
(B) E
(C) D
(D) A

Q.12

18 is related to 74 following a certain logic. Following the same logic, 22 is related to 90. To which of the following is 25 related, following the same logic? (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 102
(B) 104
(C) 100
(D) 106

Q.13

In a certain code language, ‘CASH’ is coded as ‘2486’ and ‘HIRE’ is coded as ‘3257’.What is the code for ‘H’ in the given code language?

(A) 2
(B) 6
(C) 7
(D) 3

Q.14

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: All shirts are pants. All pants are watches. Conclusions: (I) All pants are shirts. (II) All shirts are watches.

(A) Only conclusion I follows
(B) Neither conclusion I nor II follows
(C) Both conclusions I and II follow
(D) Only conclusion II follows

Q.16
In a certain code language, 'FOIL' is coded as '1974' and 'LIFE' is coded as '7291'. What is the code for 'E' in the given code language?

(A) 2
(B) 1
(C) 9
(D) 7

Q.17
Which two signs should be interchanged to make the given equation correct? 35 + 60 ÷ 12 × 4 – 12 = 203

(A) – and +
(B) + and ×
(C) – and ×
(D) ÷ and ×

Q.18
What will come in the place of '?' in the following equation, if '+' and '×' are interchanged and '–' and '÷' are interchanged? 5 + 625 – 25 × 75 ÷ 65 =?

(A) 145
(B) 135
(C) 165
(D) 125

Q.19
What should come in place of '?' in the given series? 14, 30, 93, 470, 3297, ?

(A) 29682
(B) 36267
(C) 36278
(D) 23086

General Awareness 25 questions

Q.1
Who among the following was the pioneer of 'Bhoodan Movement'?

(A) Acharya Vinoba Bhave
(B) Vallabh bhai Patel
(C) Bhimrao Ambedkar
(D) Jawaharlal Nehru

Q.2
A particular cell present in the excretory system of flatworms, which helps them removing waste materials is the:

(A) cnidoblast cell
(B) passage cell
(C) lacis cell
(D) flame cell

Q.3
The rock-cut temple of Shiva built by Krishna I is found at which of the following places?

(A) Kannauj
(B) Mahabalipuram
(C) Ellora
(D) Manyakhet

Q.4
In which of the following states is the Bhadaravi Poonam Mela organised every year in the Nilkanth Mahadev temple premises in Babra?

(A) Rajasthan
(B) Gujarat
(C) Himachal Pradesh
(D) Uttarakhand

Q.5
Which of the following cities is known as the Steel City in India?

(A) Jamshedpur
(B) Ranchi
(C) Patna
(D) Mumbai

Q.6

As per the Constitution (Ninety-First Amendment) Act, 2003, the total number of ministers, including the Prime Minister, shall NOT exceed:

(A) 25% of the total number of members of the Lok Sabha
(B) 10% of the total number of members of the Lok Sabha
(C) 15% of the total number of members of the Lok Sabha
(D) 20% of the total number of members of the Lok Sabha

Q.7

The concept of Concurrent List in the Indian Constitution is borrowed from the Constitution of _____.

(A) Britain
(B) Russia
(C) USA
(D) Australia

Q.8

In Dholavira, which of the following materials is extensively used as a building material?

(A) Bricks
(B) Granite
(C) Cement
(D) Stone

Q.9

Who became the first woman to command the Armed Forces Transfusion Centre, Delhi Cantt, which is the largest blood transfusion centre of the Armed Forces in Nov 2023?

(A) Col Shaliza Sharma
(B) Lt Col Pooja Thakur
(C) Col Sunita BS
(D) Major Anjali Bhagwat

Q.10

In which of the following year did India win the ICC T20 Men's Cricket World Cup?

(A) 2010
(B) 2012
(C) 2007
(D) 2009

Q.11

Consider the following statements in the context of the Vedas. 1.The Vedas are called Shruti. 2.The Rigveda consists of ten Mandalas. Which of the statements given above is/are correct?

(A) Both 1 and 2
(B) Only 1
(C) Neither 1 nor 2
(D) Only 2

Q.12

Which of the following government initiatives focuses on skill development to reduce open unemployment among youth in India?

(A) Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA)
(B) PM KUSUM
(C) Pradhan Mantri Kaushal Vikas Yojana (PMKVY)
(D) Swachh Bharat Abhiyan

Q.13

When was the Ustad Bismillah Khan Yuva Puraskar, an award given by the Sangeet Natak Akademi to outstanding artists under 40 who have demonstrated talent in the fields of music, dance and theatre, started?

(A) 2006
(B) 2010
(C) 2002
(D) 2008

Q.14
In a metro station, it was observed that on an average 20 people per minute were using the escalator to reach at a height of 15 metres. Calculate the power of the escalator assuming the average mass of people to be 50 kg and acceleration due to gravity to be 10 m/s^2.

(A) 2.5 kW
(B) 15 kW
(C) 150 kW
(D) 1.25 kW

Q.15
In which state are the Gua and Noamundi iron ore mines located?

(A) Karnataka
(B) Chhattisgarh
(C) Jharkhand
(D) Maharashtra

Q.16
What is the rank of Sri Lanka in the Global Hunger Index (GHI) 2023?

(A) 60
(B) 69
(C) 95
(D) 80

Q.17
Indian classical musician, Ustad Rashid Khan was awarded the Padma Bhushan in 2022 for his contribution to which of the following music categories?

(A) Folk music
(B) Carnatic music
(C) Hindustani classical music
(D) Qawwali

Q.18
In which of the following classical dance forms are the actors' faces painted to symbolise the type of character they are playing?

(A) Bharatanatyam
(B) Kathak
(C) Kathakali
(D) Odissi

Q.19
How many titles did West Bengal win in the Santosh Trophy events up to the 2022-23 season?

(A) 15
(B) 30
(C) 29
(D) 32

Q.20
The climatic diversity of India is characterised by its distinctiveness, and one of the following factors does NOT contribute to it is _____.

(A) latitudinal extent
(B) longitudinal extent
(C) topography
(D) vegetation

Q.21
What is the primary acid found in citrus fruits like oranges and lemons?

(A) Acetic acid
(B) Tartaric acid
(C) Citric acid
(D) Lactic acid

Q.22
Which Article of the Constitution of India provides provision for appointment of the same person as the Governor for two or more states?

(A) Article 157
(B) Article 153
(C) Article 158
(D) Article 158(2)

Q.23
Who among the following Governor-Generals declared 'Sati' practice as illegal?

(A) Lord Lytton
(B) Lord Canning
(C) Lord Dalhousie
(D) Lord William Bentinck

Q.24
Which of the following fictional towns is mentioned in the English novel ‘In Custody’ written by Anita Desai?

(A) Kanthapura
(B) Malgudi
(C) Lilliput
(D) Mirpore

Q.25
In 1992, which organisation launched Self Help Group – Bank Linkage Programme?

(A) SEBI
(B) SBI
(C) RBI
(D) NABARD

English Language and Comprehension 24 questions

Q.1
Select the most appropriate synonym of the word ‘Illuminate’ from the given sentence. The toughest and most typical question in the examination was to elucidate a particular theory related to friction in physics.

(A) Particular
(B) Typical
(C) Elucidate
(D) Theory

Q.2
Select the most appropriate idiom that can substitute the underlined segment in the given sentence. The manager decided to “disclose a secret” and address the issue at the team meeting.

(A) bury the hatchet
(B) hit the nail on the head
(C) spill the beans
(D) sweep it under the rug

Q.3
The following sentence contains an error. Select the option that correctly rectifies the error. If I was a king, I would have tried my best to remove poverty from my country.

(A) my best by remove poverty
(B) If I were a king,
(C) on my country
(D) I would has tried

Q.4
The given sentence has an error. Select the option that rectifies the error. The students were asked to turn up their assignments by Friday.

(A) asked turn up their assignments
(B) The students asked to
(C) their assignment by Friday
(D) turn in their assignments

Q.5
Select the most appropriate ANTONYM of the given word. Captivity

(A) Liberty
(B) Entity
(C) Authenticity
(D) Ethnicity

Q.6
The following sentence has been split into four segments. Identify the segment that contains a grammatical error. They was / celebrating their / Silver wedding anniversary / next month in Dubai.

(A) next month in Dubai
(B) celebrating their
(C) They was
(D) Silver wedding anniversary

Q.7

Select the most appropriate option that can substitute the underlined segment in the given sentence. If there is no need to substitute it, select 'No substitution required'. Everyone in the room was in "a jolly" and relaxed mood.

(A) most jolly
(B) jollier
(C) No substitution required
(D) more jollier

Q.8

Select the INCORRECTLY spelt word..

(A) Appearance
(B) Astronaut
(C) Oppertunity
(D) Immigration

Q.9

Select the most appropriate option to fill in the blank. She treats her juniors _____ contempt.

(A) by
(B) in
(C) for
(D) with

Q.10

Select the most appropriate ANTONYM of the given word. Chaotic

(A) Systematic
(B) Obscure
(C) Confused
(D) Dressed

Q.11

Select the most appropriate option that can substitute the underlined segment in the given sentence. The blanket gifted by my granny is "softer" and cozy.

(A) more soft
(B) soft
(C) softest
(D) softly

Q.12

Select the most appropriate option to fill in the blank. By next summer, they will be _____ their house.

(A) renovating
(B) renovated
(C) renovate
(D) renovates

Q.13

Select the INCORRECTLY spelt word.

(A) Illustration
(B) Occassion
(C) Efficient
(D) Boundary

Q.14

Select the most appropriate meaning of the given idiom. Out of spirits

(A) Gloomy
(B) Cheerful
(C) Obsolete
(D) Hurt

Q.15

Select the most appropriate idiom that can substitute the underlined segment in the given sentence. The new employee is really trying to fit in, but sometimes he feels "out of place and uncomfortable".

(A) a bird of a feather
(B) a diamond in the rough
(C) a needle in a haystack
(D) a fish out of water

Q.17

Select the word that has the same meaning (SYNONYM) as the underlined word in the given sentence. Almost as wide as he was tall, he had a “voracious” appetite for food and drink and a rollicking personality to match.

(A) insatiable
(B) vehement
(C) imminent
(D) venerable

Q.18

Select the most appropriate meaning of the underlined word in the given sentence. Urban people have a “bombastic” lifestyle, which is very different from those of rural folks.

(A) Pretentious
(B) Delusive
(C) Hollow
(D) Common

Q.19

Select the option that rectifies the spelling of the underlined word in the given sentence. The police informed the statesman that the goons were up to something “mischievious”.

(A) mischevous
(B) mischevious
(C) mischievous
(D) mischeivous

Q.20

Select the most appropriate option that can substitute the underlined segment in the given sentence. The unexpected guest at the party was “a bull in a China shop”.

(A) influential and meticulous
(B) careless and damaging
(C) leisurely and relaxed
(D) careful and compassionate

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. In recent years, the central and state governments in India have been taking special steps to __1__ foreign companies to invest in India. Industrial zones, called Special Economic Zones (SEZs), are being __2__. SEZs have world-class facilities, including electricity, water, roads, transport, storage, recreational, and educational facilities. Companies that set up production units in the SEZs do not have to pay taxes for an initial period of five years. The government has also __3__ flexibility in the labour laws to attract foreign investment. In recent years, the government has allowed companies to __4__ many of these provisions. Instead of hiring workers on a regular basis, companies hire workers 'flexibly' for short periods when there is __5__ pressure of work. This is done to reduce the cost of labour for the company. However, some foreign companies are not satisfied, and they are demanding more in labour laws, citing the need for greater operational efficiency.

Select the most appropriate option to fill in blank number 1.

(A) attract
(B) fulfil
(C) abhor
(D) deplore

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. In recent years, the central and state governments in India have been taking special steps to __1__ foreign companies to invest in India. Industrial zones, called Special Economic Zones (SEZs), are being __2__. SEZs have world-class facilities, including electricity, water, roads, transport, storage, recreational, and educational facilities. Companies that set up production units in the SEZs do not have to pay taxes for an initial period of five years. The government has also __3__ flexibility in the labour laws to attract foreign investment. In recent years, the government has allowed companies to __4__ many of these provisions. Instead of hiring workers on a regular basis, companies hire workers 'flexibly' for short periods when there is __5__ pressure of work. This is done to reduce the cost of labour for the company. However, some foreign companies are not satisfied, and they are demanding more in labour laws, citing the need for greater operational efficiency.

Select the most appropriate option to fill in blank number 2.

(A) wound up
(B) set up
(C) dismantled
(D) obeyed

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. In recent years, the central and state governments in India have been taking special steps to __1__ foreign companies to invest in India. Industrial zones, called Special Economic Zones (SEZs), are being __2__. SEZs have world-class facilities, including electricity, water, roads, transport, storage, recreational, and educational facilities. Companies that set up production units in the SEZs do not have to pay taxes for an initial period of five years. The government has also __3__ flexibility in the labour laws to attract foreign investment. In recent years, the government has allowed companies to __4__ many of these provisions. Instead of hiring workers on a regular basis, companies hire workers 'flexibly' for short periods when there is __5__ pressure of work. This is done to reduce the cost of labour for the company. However, some foreign companies are not satisfied, and they are demanding more in labour laws, citing the need for greater operational efficiency.

Select the most appropriate option to fill in blank number 3.

(A) allowed
(B) winked
(C) suffered
(D) invited

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. In recent years, the central and state governments in India have been taking special steps to __1__ foreign companies to invest in India. Industrial zones, called Special Economic Zones (SEZs), are being __2__. SEZs have world-class facilities, including electricity, water, roads, transport, storage, recreational, and educational facilities. Companies that set up production units in the SEZs do not have to pay taxes for an initial period of five years. The government has also __3__ flexibility in the labour laws to attract foreign investment. In recent years, the government has allowed companies to __4__ many of these provisions. Instead of hiring workers on a regular basis, companies hire workers 'flexibly' for short periods when there is __5__ pressure of work. This is done to reduce the cost of labour for the company. However, some foreign companies are not satisfied, and they are demanding more in labour laws, citing the need for greater operational efficiency.

Select the most appropriate option to fill in blank number 4.

(A) effect
(B) distort
(C) flounder
(D) ignore

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. In recent years, the central and state governments in India have been taking special steps to __1__ foreign companies to invest in India. Industrial zones, called Special Economic Zones (SEZs), are being __2__. SEZs have world-class facilities, including electricity, water, roads, transport, storage, recreational, and educational facilities. Companies that set up production units in the SEZs do not have to pay taxes for an initial period of five years. The government has also __3__ flexibility in the labour laws to attract foreign investment. In recent years, the government has allowed companies to __4__ many of these provisions. Instead of hiring workers on a regular basis, companies hire workers 'flexibly' for short periods when there is __5__ pressure of work. This is done to reduce the cost of labour for the company. However, some foreign companies are not satisfied, and they are demanding more in labour laws, citing the need for greater operational efficiency.

Select the most appropriate option to fill in blank number 5.

(A) mechanical
(B) intense
(C) artificial
(D) inadequate

Answer key

Numerical and Mathematical Ability

Q.1 **(D)**	Q.2 **(A)**	Q.3 **(D)**	Q.4 **(D)**	Q.5 **(B)**	Q.6 **(D)**
Q.9 **(A)**	Q.10 **(D)**	Q.12 **(D)**	Q.13 **(C)**	Q.14 **(A)**	Q.15 **(D)**
Q.17 **(B)**	Q.18 **(A)**	Q.19 **(C)**			

Reasoning Ability and Problem Solving

Q.1 **(D)**	Q.2 **(B)**	Q.3 **(A)**	Q.5 **(D)**	Q.6 **(C)**	Q.7 **(A)**
Q.8 **(B)**	Q.9 **(A)**	Q.10 **(D)**	Q.12 **(A)**	Q.13 **(A)**	Q.14 **(D)**
Q.16 **(A)**	Q.17 **(D)**	Q.18 **(B)**	Q.19 **(C)**		

General Awareness

Q.1 **(A)**	Q.2 **(D)**	Q.3 **(C)**	Q.4 **(B)**	Q.5 **(A)**	Q.6 **(C)**
Q.7 **(D)**	Q.8 **(D)**	Q.9 **(C)**	Q.10 **(C)**	Q.11 **(A)**	Q.12 **(C)**
Q.13 **(A)**	Q.14 **(A)**	Q.15 **(C)**	Q.16 **(A)**	Q.17 **(C)**	Q.18 **(C)**
Q.19 **(D)**	Q.20 **(B)**	Q.21 **(C)**	Q.22 **(B)**	Q.23 **(D)**	Q.24 **(D)**
Q.25 **(D)**					

English Language and Comprehension

Q.1 **(C)**	Q.2 **(C)**	Q.3 **(B)**	Q.4 **(D)**	Q.5 **(A)**	Q.6 **(C)**
Q.7 **(C)**	Q.8 **(C)**	Q.9 **(D)**	Q.10 **(A)**	Q.11 **(B)**	Q.12 **(A)**
Q.13 **(B)**	Q.14 **(A)**	Q.15 **(D)**	Q.17 **(A)**	Q.18 **(A)**	Q.19 **(C)**
Q.20 **(B)**	Q.21 **(A)**	Q.22 **(B)**	Q.23 **(A)**	Q.24 **(D)**	Q.25 **(B)**

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (D) 1,12,500

Expenditure is the other 25% of the income, so saving (75%) is three times it: 3 × ₹37,500 = ₹1,12,500.

Q.2 · Answer (A) 525

SI = 3,500 × 3 × 5/100 = ₹525.

Q.3 · Answer (D) 4 cm

The volume stays the same: $(1/3)\pi \times 7^2 \times 48 = \pi \times 14^2 \times h$, so $784\pi = 196\pi h$ and h = 4 cm.

Q.4 · Answer (D) 5.18

The cost price is 19/1.10 = ₹17.27. A 40% profit needs a selling price of 19 × 1.40/1.10 = ₹24.18, so the price must rise by ₹5.18.

Q.5 · Answer (B) 45

X costs 80/5 = ₹16 a gm and Y 80/10 = ₹8 a gm. By alligation X : Y = (14 – 8) : (16 – 14) = 3 : 1, so X = 60 × 3/4 = 45 gm.

Q.6 · Answer (D) 14.58%

The cost is 330/1.10 = ₹300. Without the 4% discount the price would be 330/0.96 = ₹343.75, a profit of 43.75/300 × 100 = 14.58%.

Q.9 · Answer (A) 9

He covers 2 km in 20 minutes, which leaves 6 km for the remaining 40 minutes (2/3 hour): 6 ÷ 2/3 = 9 km/h.

Q.10 · Answer (D) 576

$576 = 2^6 \times 3^2$, and 12, 36, 72, 144 and 288 all divide it, so the LCM is 576 itself.

Q.12 · Answer (D) 104

(50 × 105 + 5 × 95)/55 = (5,250 + 475)/55 = 5,725/55 = 104.09, or 104 cm to the nearest integer.

Q.13 · Answer (C) 27 minutes

The distance is 5.4 × 40/60 = 3.6 km; at 8 km/h it takes 3.6/8 = 0.45 hour = 27 minutes.

Q.14 · Answer (A) 48

Bowl: $(2/3)\pi \times 24^3 = 9{,}216\pi$ cm³. Each container: $\pi \times 4^2 \times 12 = 192\pi$ cm³. 9,216/192 = 48.

Q.15 · Answer (D) 525

SI = 3,500 × 5 × 3/100 = ₹525.

Q.17 · Answer (B) ₹54

(15 × 60 + 8 × 55 + 7 × 40)/30 = (900 + 440 + 280)/30 = 1,620/30 = ₹54.

Q.18 · Answer (A) 10%

On a cost of 100 the marked price is 125, and 12% off leaves 125 × 0.88 = 110: a profit of 10%.

Q.19 · Answer (C) 22 : 19

A earns 88 × 10 = ₹880 a day and B 95 × 8 = ₹760. 880 : 760 = 22 : 19.

Reasoning Ability and Problem Solving

Q.1 · Answer (D) 4

T, A and N are in both words, as are 3, 6 and 8 in both codes. The letter left in THAN is H and the digit left in its code is 4, so H = 4.

Q.2 · Answer (B) 3

A is the only letter in both words and 3 the only digit in both codes, so A = 3.

Q.3 · Answer (A) Only conclusion (II) follows
'Some plants are spices' means some spices are plants, so (II) follows. Nothing is said about lentils and the plants that are not spices, so (I) need not follow.

Q.5 · Answer (D) KNQ
The first letters move 9 places forward each time (A, J, S, B, K, wrapping round after Z), and each group is three letters 3 apart: KNQ.

Q.6 · Answer (C) XTCR
The first two letters move 4 places forward and the last two 3 back: DZRG → HDOD (Z wraps round to D) and LHLA → PLIX. So TPFU → XTCR.

Q.7 · Answer (A) LHBF
Each letter's place in the alphabet is doubled: D(4) → H(8), B(2) → D(4), C(3) → F(6), A(1) → B(2). So F(6) → L(12), D → H, A → B and C → F: LHBF.

Q.8 · Answer (B) 387
Each number is multiplied by 9: 21 × 9 = 189 and 35 × 9 = 315. Likewise 43 × 9 = 387.

Q.9 · Answer (A) T
Going round to each person's right (anticlockwise, facing the centre): G, D, T, S, R, E, F. T is second to G's right with S on T's right, E is second to G's left and R third to D's right. Third to the right of F is T.

Q.10 · Answer (D) A
Facing away from the centre, each person's right is clockwise. Going round to the right: B, E, D, A, C, F, so D is second to B's right, A is on D's right and F sits between C and B. Second to the left of F is A.

Q.12 · Answer (A) 102
Each number is multiplied by 4 and 2 is added: 18 × 4 + 2 = 74 and 22 × 4 + 2 = 90. Likewise 25 × 4 + 2 = 102.

Q.13 · Answer (A) 2
H is the only letter in both words and 2 the only digit in both codes, so H = 2.

Q.14 · Answer (D) Only conclusion II follows
All shirts are pants and all pants are watches, so all shirts are watches (II). Nothing says every pant is a shirt, so (I) does not follow.

Q.16 · Answer (A) 2
F, I and L are in both words, as are 1, 7 and 9 in both codes. The letter left in LIFE is E and the digit left in its code is 2, so E = 2.

Q.17 · Answer (D) ÷ and ×
Interchanging ÷ and ×: 35 + 60 × 12 ÷ 4 − 12 = 35 + 180 − 12 = 203.

Q.18 · Answer (B) 135
After interchanging the signs: 5 × 625 ÷ 25 + 75 − 65 = 125 + 75 − 65 = 135.

Q.19 · Answer (C) 36278
Each term is the one before times n, plus n, with n running through the primes 2, 3, 5, 7, 11: 14 × 2 + 2 = 30, 30 × 3 + 3 = 93, 93 × 5 + 5 = 470, 470 × 7 + 7 = 3297. Next: 3297 × 11 + 11 = 36278.

General Awareness

Q.1 · Answer (A) Acharya Vinoba Bhave
Vinoba Bhave began the Bhoodan (land-gift) movement in 1951 at Pochampally, in present-day Telangana, asking landowners to give land to the landless.

Q.2 · Answer (D) flame cell
Flatworms such as planarians remove wastes through flame cells, whose beating cilia (the 'flame') drive fluid into the excretory tubules.

Q.3 · Answer (C) Ellora
The Kailasa temple at Ellora, cut from a single rock, was built under the Rashtrakuta king Krishna I in the 8th century.

Q.4 · Answer (B) Gujarat
Babra is a town in Amreli district, Gujarat; 'Bhadarvi Poonam' is the full moon of Bhadarvo, the Gujarati name for the month of Bhadrapad.

Q.5 · Answer (A) Jamshedpur
Jamshedpur in Jharkhand, which grew up around the Tata steel works, is known as the Steel City of India.

Q.6 · Answer (C) 15% of the total number of members of the Lok Sabha
The 91st Amendment added Article 75(1A): the Council of Ministers, Prime Minister included, may not exceed 15% of the total number of members of the Lok Sabha.

Q.7 · Answer (D) Australia
The Concurrent List, like freedom of trade and commerce and the joint sitting of Parliament, was borrowed from the Australian Constitution.

Q.8 · Answer (D) Stone
Unlike most Harappan cities, which were built of brick, Dholavira in the Rann of Kutch was built largely of stone.

Q.9 · Answer (C) Col Sunita BS
Colonel Sunita B.S. took command of the Armed Forces Transfusion Centre, Delhi Cantt, in November 2023, the first woman to do so.

Q.10 · Answer (C) 2007
India won the first ICC T20 World Cup, in South Africa in 2007, beating Pakistan in the final; its second title came in 2024.

Q.11 · Answer (A) Both 1 and 2
The Vedas are 'Shruti', what was heard or revealed, as against 'Smriti', what is remembered; the Rigveda's hymns are arranged in ten books, the Mandalas.

Q.12 · Answer (C) Pradhan Mantri Kaushal Vikas Yojana (PMKVY)
PMKVY, launched in 2015, trains young people in job skills. MGNREGA guaranteed wage work when the exam was held (October 2024), PM KUSUM promotes solar pumps for farmers and Swachh Bharat is about sanitation.

Q.13 · Answer (A) 2006
The Sangeet Natak Akademi instituted the Ustad Bismillah Khan Yuva Puraskar in 2006.

Q.14 · Answer (A) 2.5 kW
Power is the work done (mgh) per second: (20 × 50 × 10 × 15)/60 = 2,500 W = 2.5 kW.

Q.15 · Answer (C) Jharkhand
Gua and Noamundi are iron-ore mines in the West Singhbhum district of Jharkhand.

Q.16 · Answer (A) 60
Sri Lanka ranked 60th of 125 countries in the Global Hunger Index 2023; India was 111th.

Q.17 · Answer (C) Hindustani classical music
Ustad Rashid Khan, of the Rampur-Sahaswan gharana, was a Hindustani classical vocalist; he received the Padma Bhushan in 2022.

Q.18 · Answer (C) Kathakali
Kathakali, Kerala's dance-drama, paints the actors' faces in set colours for each type of character, such as green (pacha) for noble heroes.

Q.19 · Answer (D) 32
Up to the 2022-23 season Bengal had won the Santosh Trophy, the national football championship for state teams, a record 32 times, the last in 2016-17.

Q.20 · Answer (B) longitudinal extent
Longitude decides a place's local time, not its climate. Latitude and relief (topography) shape temperature, winds and rainfall, and vegetation affects local humidity and temperature.

Q.21 · Answer (C) Citric acid
Citrus fruits owe their sour taste mainly to citric acid. Tartaric acid is found in grapes and tamarind, lactic acid in curd and acetic acid in vinegar.

Q.22 · Answer (B) Article 153
Article 153 provides a Governor for each State, and its proviso, added by the Seventh Amendment (1956), allows the same person to be Governor of two or more States.

Q.23 · Answer (D) Lord William Bentinck
Lord William Bentinck's Bengal Sati Regulation of 1829 made sati illegal, with the support of Raja Ram Mohan Roy.

Q.24 · Answer (D) Mirpore
Anita Desai's 'In Custody' (1984) is set in Mirpore, where the Hindi lecturer Deven teaches. Kanthapura is Raja Rao's village, Malgudi R. K. Narayan's town and Lilliput Swift's island.

Q.25 · Answer (D) NABARD
NABARD launched the SHG-Bank Linkage Programme in 1992, linking self-help groups of the poor to bank credit.

English Language and Comprehension

Q.1 · Answer (C) Elucidate
To 'elucidate' is to make clear or explain: to 'illuminate'.

Q.2 · Answer (C) spill the beans
To 'spill the beans' is to reveal a secret.

Q.3 · Answer (B) If I were a king,
An imagined condition takes the subjunctive 'were' with every subject: 'If I were a king'.

Q.4 · Answer (D) turn in their assignments
To 'turn in' work is to hand it in; to 'turn up' is to arrive or appear.

Q.5 · Answer (A) Liberty
'Captivity' is being held prisoner; its opposite is 'liberty', freedom.

Q.6 · Answer (C) They was
The plural 'they' takes 'were': 'They were celebrating'.

Q.7 · Answer (C) No substitution required
Nothing is being compared, so the positive degree 'a jolly and relaxed mood' is right; 'more jollier' is a double comparative.

Q.8 · Answer (C) Oppertunity
The correct spelling is 'opportunity'.

Q.9 · Answer (D) with
One treats someone 'with' contempt.

Q.10 · Answer (A) Systematic
'Chaotic' means in complete disorder; its opposite is 'systematic'.

Q.11 · Answer (B) soft
Nothing is being compared, and the adjective must match 'cozy' in degree: 'soft and cozy'.

Q.12 · Answer (A) renovating
'Will be' with the -ing form makes the future continuous: 'they will be renovating their house'.

Q.13 · Answer (B) Occassion
The correct spelling is 'occasion', with one 's'.

Q.14 · Answer (A) Gloomy
To be 'out of spirits' is to be low or sad: gloomy.

Q.15 · Answer (D) a fish out of water
'A fish out of water' is someone uncomfortable in unfamiliar surroundings.

Q.17 · Answer (A) insatiable
'Voracious' means wanting great quantities and never satisfied: 'insatiable'.

Q.18 · Answer (A) Pretentious
'Bombastic' means pompous and inflated, high-sounding but with little substance: 'pretentious'.

Q.19 · Answer (C) mischievous
There is no 'i' after the 'v': 'mischievous'.

Q.20 · Answer (B) careless and damaging
'A bull in a china shop' is someone clumsy and careless who causes damage.

Q.21 · Answer (A) attract
The governments want to 'attract' foreign companies to invest, as the passage says again later ('to attract foreign investment').

Q.22 · Answer (B) set up
SEZs with world-class facilities are being 'set up', that is established; 'wound up' and 'dismantled' mean the opposite.

Q.23 · Answer (A) allowed
The government has 'allowed' flexibility in the labour laws to attract foreign investment.

Q.24 · Answer (D) ignore
Companies have been allowed to 'ignore' these provisions, hiring workers flexibly instead of on a regular basis.

Q.25 · Answer (B) intense
Workers are hired for short periods when there is 'intense', that is heavy, pressure of work.

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